

RESEARCH ARTICLE

Concrete crack segmentation based on convolution–deconvolution feature fusion with holistically nested networks

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Summary

Automatic crack detection on concrete surfaces has become increasingly important for the health diagnosis of concrete structures to prevent possible malfunctions or accidents. In this paper, a concrete crack segmentation network based on convolution–deconvolution feature fusion with holistically nested networks is proposed. The proposed network adopts an encoder–decoder structure and uses VGG-16 as the basic feature extraction network. First, considering the problem that the VGG-16 network can extract redundant features in the encoding stage, based on the channel attention mechanism, the channel spatial correlation and global information are used to emphasize crack features to remove redundant features. Second, through the convolution–deconvolution feature fusion module, the deep semantic information of the deconvolution is effectively fused with the shallow features of convolution, which effectively improves the semantic crack feature information extracted at each stage of the VGG-16 network. Finally, based on a multiscale supervised learning mechanism, holistically nested networks are used to fuse the prediction results from different scales, which enhances the network's ability to express linear topological structures and improves the accuracy of crack segmentation. Through a large number of experiments on the Bridge_Crack_Image_Data dataset and CFD dataset, we demonstrate that compared with other deep networks, the proposed network not only achieves better segmentation results for cracks of different widths but is also more robust.

KEYWORDS

channel attention mechanism, convolution–deconvolution feature fusion, crack segmentation, holistically nested networks, VGG-16 network

1 | INTRODUCTION

With the rapid development of civil infrastructure, many essential critical infrastructure components, such as bridges, dams, highways, and large public buildings, have been built. During the long-term operation of infrastructure, damage to building structures is caused by the coupled effects of many factors, such as design, construction, climate, and

environment. Cracks on concrete structure surfaces are critical indicators of structural damage and durability.^{1–4} Crack detection plays an important role in the structural health of concrete and reliability maintenance of infrastructure. It is necessary to conduct regular inspections and maintenance of infrastructure to reduce the risks of structural deterioration. Thus, timely and accurate crack detection on such surfaces is of great significance for targeted remediation of concrete building structures, emergency treatments, and for preventing safety-related accidents.

Early concrete crack detection was performed mainly through manual visual inspection; however, such inspections can be time-consuming, laborious, expensive, and/or dangerous.⁵ As computer vision technology has developed, vision-based concrete crack detection applications have attracted increasing attention from scholars in recent years due to their low cost, ease of development, and rapid processing. However, due to the complex characteristics of building images collected in natural scenes, vision-based concrete crack detection remains a challenging task. Many vision-based approaches have been proposed to solve this challenge. These approaches can be divided into traditional and deep learning-based approaches according to the different computer vision methods applied.

The traditional approaches can be grouped into three categories: thresholding-based approaches, feature-based approaches, and edge-based approaches. In early studies, thresholding-based approaches^{6–11} were often proposed to detect cracks. However, choosing an appropriate threshold is difficult. In addition, these approaches are sensitive to different types of disturbances, such as noise, shadows, and unexpected light areas, resulting in poor performance. From the feature-based approach perspective, Oliveira utilized a one-class classification strategy to separate crack and non-crack regions via the mean and variance features of image regions. However, the use of these simple features alone presents difficulty to achieving good detection results in situations with complex backgrounds. Bray et al.¹² computed histogram features and input them into a neural network to classify images with and without cracks. Xu et al.¹³ proposed an asphalt pavement crack recognition algorithm based on histogram estimation and shape analysis. Hu and Zhao¹⁴ developed a local binary pattern (LBP) operator with a lookup table to classify crack and noncrack pixels for pavement surfaces. However, when more complex crack features are used, noisy patterns might still be falsely detected.¹³ To address the low contrast of tiny cracks and noisy patterns, Chen et al.¹⁵ proposed an LBP-SVM crack detection approach, which adopted an LBP descriptor to extract texture characteristics, and used an SVM classifier to detect cracks on complex surfaces and noisy backgrounds. Zalama et al.¹⁶ extracted features from Gabor filtered images and utilized AdaBoost to construct strong classifiers to detect pavement cracks. Bu et al.¹⁷ employed wavelet-based image features and an SVM to detect cracks in complex bridge images. Chun et al.¹⁸ proposed a methodology that employs a light gradient boosting machine (LightGBM) and considered features based on the pixel values and geometric shapes of cracks. As the above studies show, approaches integrating more sophisticated feature descriptors and machine learning techniques could improve the accuracy of detecting cracks. The intensity values of concrete crack regions can exhibit sharp changes over short distances that can visually be clearly perceived as separated boundaries between crack and background pixels.^{19,20} Therefore, many edge-based concrete crack detection methods, such as the fast Harr transform (FHT),²⁰ fast Fourier transform,²⁰ Sobel edge detector,^{21,22} and Canny edge detector^{23–25} have been proposed. Furthermore, similar to edge-based crack detection methods, Wang et al.²⁶ proposed an integrated algorithm combining the Prewitt edge detector with the Otsu-based method for segmenting crack regions from the background. Wang et al.²⁷ utilized the K-means clustering algorithm to exploit crack regions based on rough Canny detection results. Other methods, such as the Gabor filter^{14,28} and morphology,^{10,29} have achieved some successful detection results. Nevertheless, edge detection is an ill-posed problem because image derivatives are easily affected by various types of noise.³⁰ Thus, the existing edge-based concrete crack detection methods are still sensitive to strong noise corruption.

Traditional image processing approaches^{8,31–34} often utilize manually designed concrete features to detect concrete cracks. However, it is difficult for manually designed feature expression methods used in traditional approaches to effectively describe the concrete crack features in complex natural scenes. Deep learning has demonstrated powerful feature representation capabilities in a variety of applications. Based on deep learning theory, scholars worldwide have performed extensive and in-depth research in the concrete building image crack detection field.

In early research on crack detection based on deep learning,^{35–37} researchers first divided concrete building images into subblocks and then marked the subblocks based on binary classification detection performed with a convolutional neural network to determine whether cracks existed in the subblocks. However, this method cannot quantitatively evaluate the size of cracks and is prone to missed and false detection.

In recent years, deep convolutional neural networks³⁸ have increasingly been applied to the task of semantic image segmentation, and the accuracy and universality of semantic segmentation algorithms have been greatly improved. However, for concrete crack images, the CNN-based semantic segmentation method often results in inaccurate segmentation results for the local details of the image as the dimensionality and resolution decrease, making it difficult to

obtain accurate concrete crack segmentation results.³⁹ Fully convolutional networks (FCNs) can obtain pixel-level semantic segmentation results of images; therefore, these networks have received significant attention in the concrete crack detection field. To ameliorate the slow speeds of CNN-based networks and reduce the number of redundant computations, Yang et al.⁴⁰ utilized convolutional layers to replace all the fully connected layers in the traditional CNN network and then built a novel FCN to achieve split-pixel-level segmentation. Wang et al.⁴¹ built a crack FCN model based on the FCN model to address the problems that FCNs lose local information and the capacity of partial refinement. The proposed network removed the dropout method from the fully connected layers and progressively deepened the depth of the FCN network. Huyan et al.⁴² proposed a convolutional–deconvolutional-based “U”-shaped deep neural network architecture for pixelwise pavement crack detection. Transpose convolution layers and concatenation operations were adopted to better utilize the feature information and improve the network training performance. Ren et al.⁴³ proposed a new end-to-end crack detection method based on an encoder–decoder architecture that used a spatial pyramid pooling module to obtain multiscale contextual information and improve the accuracy of crack segmentation. Cao et al.⁴⁴ proposed an attention-based crack detection network (ACNet), which includes an attention-based feature extraction module to capture the global information and an attention-based decoder module that removed redundant information. In addition, to capture the linear and structural information of cracks, some works have incorporated multiscale modules into the deep network. Qin et al.⁴⁵ proposed the DeepCrack network, in which multiscale deep features learned during the hierarchical convolutional stages are fused to capture the linear structures. Yang et al.⁴⁶ built a feature pyramid and a hierarchical boosting network for pavement crack detection. The feature pyramid integrated contextual information from high-level layers to enrich low-level features, and the hierarchical boosting network used reweighted samples to hierarchically process more difficult examples. Moreover, multiscale supervised learning was adopted to improve the network’s ability to capture the linear topology of cracks.

Inspired by the above studies, and aiming at the problem that traditional crack image segmentation methods cannot accurately extract cracks, we propose convolution–deconvolution feature fusion with holistically nested networks (CDFHNet) for concrete crack detection. The main contributions of this paper are as follows:

1. Aiming at the problem of crack feature redundancy in encoding–decoding architecture, a channel attention mechanism is constructed to improve the attention of the VGG-16 feature extraction network to the original feature information of cracks. We establish an adaptive excitation method that can evaluate the importance of each channel of the original feature map to weight and output the channel features according to importance. Prescreening VGG-16 feature extraction enhances the effective feature responses of each scale in the feature map and weakens invalid feature responses.
2. The convolution–deconvolution feature fusion module introduces the shallow features extracted by the VGG-16 network into the deep semantic information for feature fusion and enhances the semantic crack feature information extracted at each stage of the VGG-16 network. This approach eliminates interference such as noise and shallow-feature background textures, allowing more significant concrete crack characteristics to be obtained at different scales.
3. We proposed a holistically nested network that integrates multiscale information. Based on the deep supervision (DS) network, supervised learning is conducted on the fusion features extracted by the feature fusion module at each stage, which enhances the feature response to the crack region through the fusion of the multiscale features between layers while weakening the feature response to noncrack regions, thereby effectively improving the network’s response to the linear topological structures of cracks and increasing the accuracy of concrete crack detection.

The content of this paper is organized as follows. Section 2 presents the details of the proposed crack segmentation network architecture, including the VGG-16 feature extraction module based on the channel attention mechanism, the convolution–deconvolution feature fusion module, and the multiscale feature fusion holistically nested network. Section 3 describes the proposed network loss function. Experimental results and demonstrations are reported and analyzed in Section 4. Finally, conclusions are presented in Section 5.

2 | OVERVIEW OF PROPOSED METHOD

Because of the variety of noise and the complexity and diversity of background textures in concrete crack images, manually designed feature expression methods used in traditional approaches have difficulty effectively describing concrete

crack features, making crack detection difficult. The powerful feature extraction ability of deep networks can capture complex concrete crack features—even linear topological structural features embedded in concrete cracks. Based on the above, we propose a convolution–deconvolution feature fusion holistically nested network for concrete crack detection. The proposed network architecture is shown in Figure 1 and is composed of three main components: (a) The VGG-16 feature extraction module fused with the channel attention mechanism. (b) The convolution–deconvolution feature fusion module. (c) The multiscale feature fusion holistically nested network.

2.1 | VGG-16 feature extraction based on the channel attention mechanism

A classic deep learning network consists of several convolutional layers, pooling layers and fully connected layers. The network first constructs a feature map through the convolution and pooling layers, then reduces the feature map into a one-dimensional vector to form the input of the fully connected layer, and finally uses the loss function to supervise the entire training process. On this basis, Simonyan and Zisserman⁴⁷ explored the relationship between network depth and its performance and constructed an efficient feature extraction network structure—the VGG-16 network. In particular, the overall structure of the VGG-16 network includes a five-stage convolutional block and three fully connected layers. The numbers of convolutional layers in each stage are 2, 2, 3, 3, and 3. The max pooling layer is included at the end of each convolution stage. Finally, three fully connected layers are used to reduce the dimensionality of deep features, and the prediction results are output according to the loss function. To achieve end-to-end pixel-level segmentation prediction, this paper removes all fully connected layers and only uses the VGG-16 convolutional network as a feature extractor. For an input image of size $M \times M$, the parameters of each convolutional stage and the size of the output feature map after pooling are shown in Table 1.

However, the convolutional stage in VGG-16 can only fuse the spatial and channel information in the local receptive field. It does not consider the global correlation between channels and thus ignores the contextual relationship outside the receptive field. Therefore, the ability of the VGG-16 network to express global information is limited. Using VGG-16 as the fracture feature extraction network, part of the channel feature visualization diagram is shown in

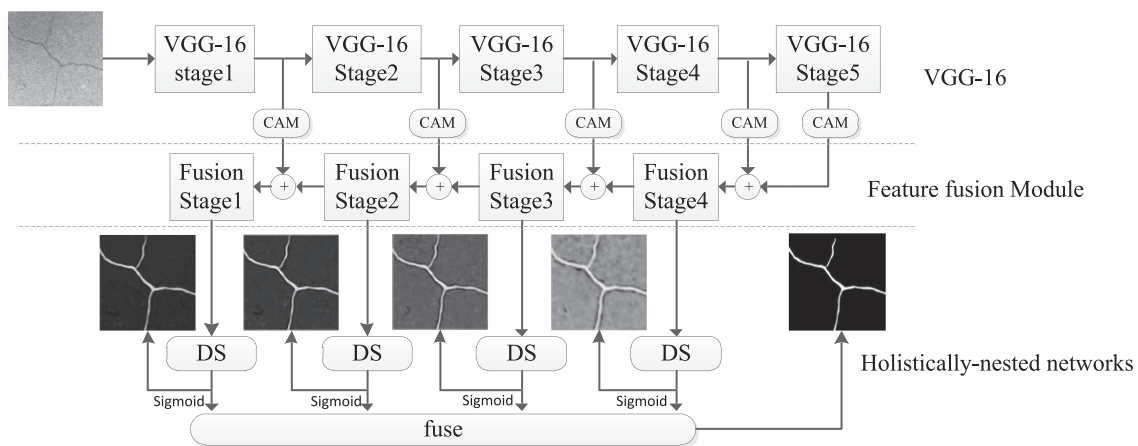


FIGURE 1 CDFFHNet network structure

TABLE 1 VGG-16 network structure parameters

Stage	Convolution	Number of channels	Pooling	Output
1(1_1,1_2)	3×3	64	Max, 2×2	$M \times M$
2(2_1,2_2)	3×3	128	Max, 2×2	$M/2 \times M/2$
3(3_1,3_2,3_3)	3×3	256	Max, 2×2	$M/4 \times M/4$
4(4_1,4_2,4_3)	3×3	512	Max, 2×2	$M/8 \times M/8$
5(5_1,5_2,5_3)	3×3	512	Max, 2×2	$M/16 \times M/16$

Figure 2. By visualizing the feature map response of each VGG-16 channel, nearly 1/3 of the feature responses are revealed to be either incorrect or invalid. As show in Figure 2, the first three columns' features are valid, and the last four columns are not. These responses are not conducive to the final network classification or the semantic segmentation of crack pixels. In addition, we found that the semantic information contained in feature channels has strong correlation. Using this, we can selectively enhance highly activated channels and inhibit invalid channels, thereby significantly improving the feature expression ability of the network. In this way, this paper improves the VGG-16 network feature extraction module by introducing a channel attention module (CAM) at the end of each stage. This allows the network to learn the global correlation between the feature channels, thereby enhancing the network's spatial expression ability. Specifically, the crack features extracted by VGG-16 are prescreened so that the feature extraction network can selectively enhance the channel feature response that contains crack information and suppress useless channel feature responses that do not reflect or only weakly reflect crack information, thus solving the problem that the traditional VGG-16 network can only learn features from the local receptive field.

Let $\mathbf{Y} \in \mathbb{R}^{C \times H \times W}$ denote the input feature map of the VGG-16 network, where represents the real number space, $C \times H \times W$ represents the dimension of the input feature $\mathbf{Y}$, where C is the channel number of the feature map $\mathbf{Y}$, and H and W are the height and width of the feature map, respectively. The adopted CAM structure is shown in Figure 3. First, a squeeze process is established for the input feature map $\mathbf{Y}$. That is, a global average pooling operation is conducted on the two-dimensional spatial area ($H \times W$) of the feature map $\mathbf{Y}$ to extract the global average response of each channel. The global spatial features of each channel are compressed to form the channel representation $D = \{D_c | c = 1, 2, \dots, C\}$, whose dimension is $C \times 1 \times 1$. The squeeze process is expressed as follows:

$$D_c = S(Y_c) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W y_c(i, j), \quad (1)$$

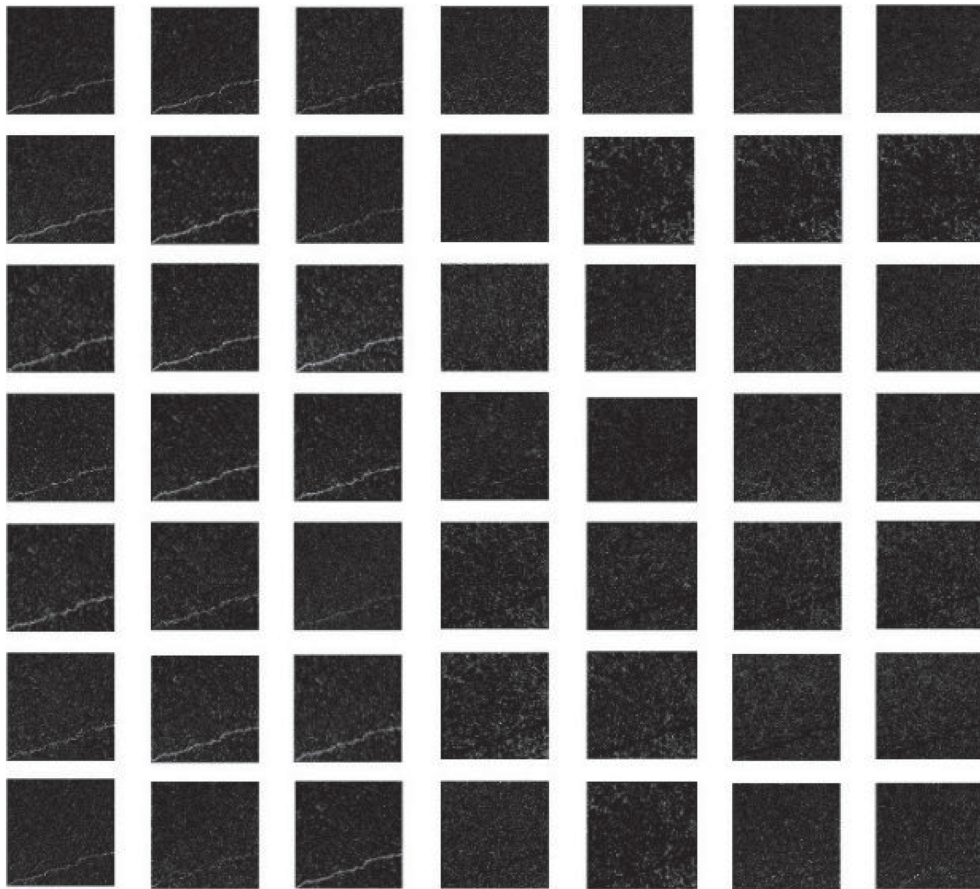


FIGURE 2 Partial channel feature visualization of stage 1 in VGG-16

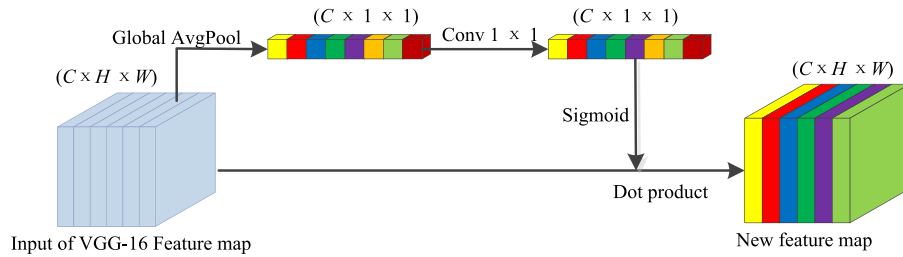


FIGURE 3 Channel attention module (CAM)

where D_c represents the c th channel descriptor and $S(Y_c)$ represents the squeeze process of feature map Y_c , where Y_c represents the c th network input feature, $c = 1, 2, \dots, C$ represents the number of channels, and $y_c(i, j)$ represents the element value in the i th row and j th column of the c th network input feature.

The channel representation D efficiently describes channel-wise correlation by embedding the global distribution of the channel feature response. By modeling the responsivity of each channel, the network's representation ability can be improved, and the features can be adjusted channel by channel, which means that the network can learn to use global information to selectively enhance the features containing useful crack information and suppress the useless features.

We then perform the excitation process to fully capture channel-wise dependencies and use the $C \times 1 \times 1$ dimensional feature vector $D = \{D_c | c = 1, 2, \dots, C\}$ of each channel as the channel global feature. Based on the principle of a simple self-gating mechanism, an extraction descriptor T is generated by learning the channel global features through a 1×1 convolutional layer, and the output dimension is still $C \times 1 \times 1$. The extraction descriptor T for the different channels is obtained using the sigmoid function, so the excitation process fully captures the dependence of each channel and controls the characteristic excitation of each channel. The excitation process is expressed as follows:

$$T = E(D, W) = \delta(g(D, W)) = \sigma(W_2 \delta(W_1 D)), \quad (2)$$

where T denotes the feature descriptor, $E(D, W)$ represents the excitation process, $\delta(\cdot)$ is the ReLU function, $\sigma(x) = 1/(1 + e^{-x})$ is a sigmoid function, and $W_1 \in \mathbb{R}^{r \times C}$ and $W_2 \in \mathbb{R}^{C \times r}$, where $r = 16$ denotes a hyperparameter.

The extraction descriptor T for the different channels is the weight distribution of the channels. Equation 2 shows that the excitation process parameterizes the gating mechanism to fit the complex correlation between the feature channels by forming a bottleneck with two fully connected (FC) layers, which include a dimensionality-reduction layer with parameters W_1 and a dimensionality-increasing layer with parameters W_2 . By training these parameters W_1 and W_2 with a loss function, we can obtain a one-dimensional excitation weight to activate each layer of channels. Therefore, the weight represents the correlation between the channel and the key information: the greater the weight, the greater the correlation.

Finally, the learned channel global feature extraction descriptor T and VGG-16 network input feature Y are subjected to a matrix point multiplication operation for the corresponding channel, and the channel information of the VGG-16 feature map is excited and screened, after which it explicitly expresses the global spatial correlation between feature channels, thereby improving the responses of effective crack features and weakening the responses of invalid features. After the input feature Y of the VGG-16 network is re-excited to the original feature in the channel dimension through the CAM, the obtained output feature map S is rescaled to the transformation output Y with the activations:

$$S = F_{scale}(T, Y) = \{T_c \cdot Y_c\}_{c \in C}, \quad (3)$$

where $F_{scale}(T, Y)$ refers to channel-wise multiplication between the weight parameter T and the input feature Y , $T_c \in T$ is the weight parameter of the c th channel obtained through global average pooling learning, $Y_c \in Y$ denotes the feature of the c th channel of the input feature map of the VGG-16 network, T is the set of weight parameters for each channel, and Y is the feature set for each channel.

The activations are actually a process of rescaling and weighting the previously obtained normalized weights T to the features of each channel. Multiplying the values of different channels with different weights can enhance the

attention to the key channel domains. In this regard, adjusting the weight of each channel feature according to the input data helps to enhance the distinguishability of the feature.

2.2 | Layer feature fusion method based on a convolution–deconvolution network

Different layers of deep convolutional neural networks encode different-level information. High-layer features always contain more semantic information, and low-layer features contain more detailed information.^{48,49} That is, the lower layers of the network can capture more spatial detail information from the image, while the higher layer encodes rich semantic information. In particular, stages 1–3 of the network contain more detailed crack information, providing effective information for detecting an accurate crack position, but the lower layer features lack sufficient semantic information. Stages 4–5 mainly contain crack semantic features, but more crack detail feature information is lost. However, detailed features and semantic features are very important for accurate crack segmentation. Therefore, we fuse the features from different scales so that the output is integrated into the detailed features of the lower layer cracks. This approach effectively improves the accuracy of crack detection. However, because concrete crack images contain interference from complex feature information, such as background textures and various types of noise, stages 1–3 of the VGG-16 network not only extract many detailed features of cracks but also extract many features that represent undesirable complex background textures, noise and other types of interference. These interference features bring great difficulties to the accurate segmentation of crack details.

Therefore, to solve the mis-segmentations caused by strong interference, such as complex background texture and noise, we propose a feature fusion method based on a convolution–deconvolution learning operation. The proposed method first includes the rich semantic feature maps from stage 5 of the VGG-16 network, then uses deconvolution learning to automatically extract the high-level semantic features of crack images, and finally reconstructs the deconvolution features from stage 1 to stage 5. The deconvolution features at each reconstruction stage not only possess more robust features containing high-level crack semantic information but also effectively eliminate other strong interference features, such as complex background textures and noise. The reconstructed stage 1 to stage 5 deconvolution features and the features extracted by the corresponding level VGG-16 network are sent to the feature fusion module to perform multiscale feature splicing and fusion. The convolution–deconvolution feature fusion method uses 1×1 convolutions to fuse the convolution–deconvolution features and transfers the high-level semantic feature information to the low-level semantic feature information, enhancing the high-level semantic features of the crack image while retaining the detail features and effectively eliminating the interference from complex background texture and noise features. For image analysis and synthesis, a more robust image feature representation can be learned by constructing a hierarchical deconvolution network.⁵⁰ This hierarchical deconvolution network structure helps the network extract the intermediate and advanced feature representations from the image.

The convolution–deconvolution feature fusion module structure is shown in Figure 4. The red box represents the feature extracted by VGG-16, and the green box denotes the reconstructed deconvolution feature. The two features are spliced and fused at a ratio of 1:1, and the fusion feature map is learned through the 1×1 convolutions. The parameters at each convolutional stage and the size of the output feature map after pooling are shown in Table 2. This process can be expressed as follows:

$$O_s = \text{concatenate}(I_c^s, I_t^s) * K_s \quad (4)$$

where O_s denotes the splicing and fusion output features at the s th stage; I_c^s represents the features of the s th stage in the c th channel extracted by the VGG-16 network; I_t^s represents the features of the s th stage reconstruction based on the deconvolutional network; $\text{concatenate}(I_c^s, I_t^s)$ represents the channel splicing and fusion operation on the extracted I_c^s, I_t^s feature maps; K_s is a 1×1 convolution kernel; and “ $*$ ” denotes the convolution operation.

Compared with the features extracted by the VGG-16 network, the proposed interlayer feature fusion network based on the convolution–deconvolution network uses deconvolution learning to capture higher level semantic features of crack images and merges high-level semantic features with low-level features layer by layer. The VGG-16 network extracts the five-stage crack features for semantic reconstruction and realizes the multistage, cross-channel, feature information interaction and fusion of the VGG-16 network—especially for lower stage features such as those of stages 1–3, which contain less semantic information. The proposed feature fusion network allows the lower-stage features to

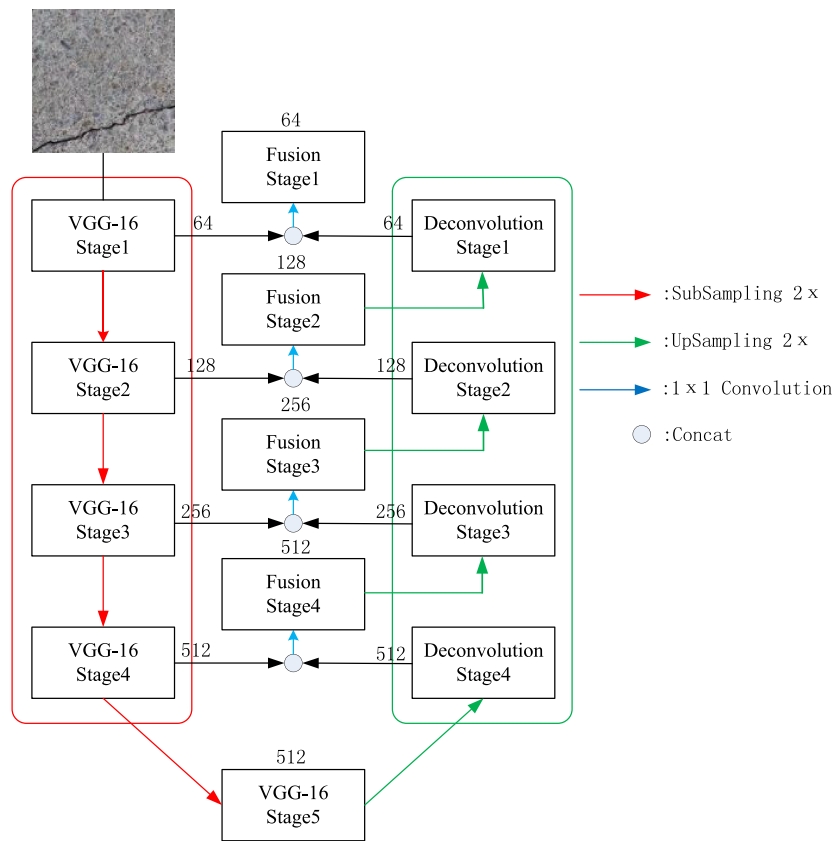


FIGURE 4 Convolution-deconvolution feature fusion module

TABLE 2 Deconvolution network structure parameters

Deconvolution stage	Convolution	Number of channels	Pooling	Output
convolution stage5_3	/	512	Max, 2×2	$2M \times 2M$
Fusion with convolution stage 4_3	1×1	512	/	$2M \times 2M$
D4(4_1,4_2,4_3)	3×3	256	Max, 2×2	$4M \times 4M$
Fusion with convolution stage 3_3	1×1	256	/	$4M \times 4M$
D3(3_1,3_2,3_3)	3×3	128	Max, 2×2	$8M \times 8M$
Fusion with convolution stage 2_2	1×1	128	/	$8M \times 8M$
D2(2_1,2_2)	3×3	64	Max, 2×2	$16M \times 16M$
Fusion with convolution stage 1_2	1×1	64	/	$16M \times 16M$
D1(5_1,5_2,5_3)	3×3	64	/	$16M \times 16M$

Note: Parameters: D denotes a deconvolution, no D denotes a convolution, and M is the initial image size.

not only retain the rich crack details but also integrate more middle- and high-level semantic features of cracks, effectively eliminating interference from complex background textures, noise, and so forth.

2.3 | Holistically nested network based on convolution-deconvolution feature fusion

The interlayer feature fusion method based on the convolution-deconvolution network combines the features of concrete cracks of different scales in the VGG-16 network and effectively reconstructs the multiscale features of concrete cracks. However, these reconstructed multiscale features still contain considerable redundant information. Therefore,

simply fusing these multiscale features can easily lead to the loss of key information in the higher layer side-output. This not only fails to effectively improve the effect of concrete crack segmentation but also requires large amounts of redundant calculations. Concrete cracks have similar regularity at different scales and simultaneously have linear topological features as a whole. Therefore, different network scales can be used to distinguish “cracks.” If multiple scales indicate that a pixel is part of a “crack,” then that point is more likely to actually be part of a “crack”; conversely, if multiple scales indicate that the pixel is not part of a “crack,” that point is more likely to be part of the background. Based on this strategy, to improve the network’s ability to describe the overall linear topology features of concrete cracks and reduce the feature redundancy problem, we propose a holistically nested network that integrates contextual information. The proposed network uses deep supervision (DS) to automatically perform nested multiscale feature learning on the convolutional features of cracks at different scales to enhance the response of crack features. At the same time, based on end-to-end network learning, the side output of different scales is introduced to stimulate a continuous response at each scale to improve the network’s response to the linear crack topology of concrete. Unlike a classic deep learning network that only produces a one-sided prediction, the holistically nested network that integrates contextual information has four side networks, which stimulate edge responses at different scales and then merge the segmentation results for each scale to obtain a final concrete crack image segmentation result. The proposed holistically nested network structure is shown in Figure 5. The network input is an extracted concrete crack image fusion module containing the overall features O of different scales:

$$O = \{O_i \mid i = 1, 2, 3, 4\}, \quad (5)$$

where O_i denotes the characteristic response of the i th side network output.

Since supervised learning of hidden layer features can effectively improve the optimization and generalization of image classification tasks,⁵¹ we propose to use the DS module to supervise the learning of the fusion features at each stage and obtain the concrete crack image segmentation results at the corresponding scale. The DS module structure is shown in Figure 6. The proposed holistically nested network sends the fusion features O obtained from the interlayer feature fusion module of the convolution–deconvolution network to the DS module for relearning, and then the DS module uses a 1×1 convolution to fuse the features and reduce the dimensionality. After dimensionality reduction, a one-channel feature and its feature response activation map are obtained. Finally, the scale of the original feature map

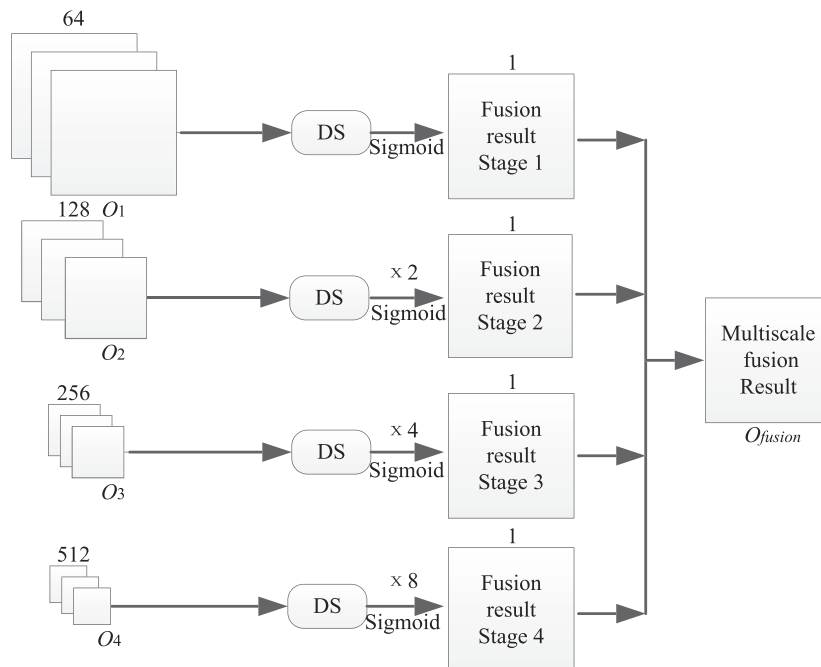


FIGURE 5 Holistically nested network



FIGURE 6 Deep supervision module

is restored through the deconvolution operation, and subsequent feature stitching is performed. The feature response activation maps of different scales are merged by the 1×1 convolution, which effectively enhances the feature responses to crack areas and weakens the feature responses to noncrack areas, thereby effectively improving the response of the holistically nested network to the linear topology of concrete cracks. The feature fusion process of the network fusion layer can be expressed as follows:

$$O_{fusion} = w_f * O = w_f * \{O_1, O_2, O_3, O_4\}, \quad (6)$$

where O_{fusion} denotes the output feature of the side network after fusion; w_f indicates the weight of the corresponding scale feature fusion layer, which contains $64 \ 1 \times 1$ convolution kernels. The notation $\{O_1, O_2, O_3, O_4\}$ represents the feature set of concrete crack images at different scales extracted by each stage, and “ $*$ ” represents the convolution operation.

3 | NETWORK LOSS FUNCTION

The network loss function measures the difference between the input training image and the output network prediction value. During training of the neural network, the network uses this error to backpropagate and iteratively optimize the network weights. When the loss function metric value reaches its smallest value, the difference between the input training image and the output network prediction value is the smallest, and the network is optimally trained at this time. The purpose of network training is to use the proposed holistically nested network to output the predicted crack mapping result according to an input crack segmentation truth map. The network loss function consists of two parts: side network loss and fusion layer loss.

3.1 | Side network loss

In the proposed holistically nested network, each side network is connected to a sigmoid classifier, the output of each layer is predicted according to the network fusion feature, and the weighted summation of the multiple side network prediction values is performed. The total loss function $L_{side}(X, Y, W, w)$ of the side network is defined as follows:

$$L_{side}(X, Y, W, w) = \sum_{m=1}^M l_{side}^{(m)}(X, Y, W, w^{(m)}), \quad (7)$$

where $L_{side}(\cdot)$ denotes the total loss function of the side network; $l_{side}^{(m)}(\cdot)$ represents the loss function of the m th side network; $X = \{X^1, \dots, X^N\}$ is a training set; $Y = \{Y^1, \dots, Y^N\}$ represents the set of truth maps corresponding to the training set; W is a set of standard parameters for the entire network; $w = (w^{(1)}, \dots, w^{(M)})$ represents the corresponding layer connection weight parameter set, where $w^{(m)}$ is a weight parameter for the m th side network; and M is the total number of output networks on the side of the holistically nested network.

During network training, the total loss of the side network comes from the error of the concrete crack image training set X and its corresponding crack segmentation truth map image set Y . However, when the positive and negative samples of the training set are balanced, the conventional loss calculation is suitable. However, in real-world concrete crack images, more than 90% of the regions are noncrack areas. Therefore, the image feature ratios of crack pixels and noncrack pixels are severely unbalanced. This imbalance leads to the problem that when the training results are used

for prediction, the training network tends to regard cracks as negative. If the samples are used for prediction, the network may learn more noncrack features than expected, which leads the network to predict more noncrack features. To solve this problem caused by class imbalance in the crack image data, we propose a new loss function based on a holistically nested network. The proposed function assigns corresponding weights to crack pixels and noncrack pixels according to the ratio of crack to noncrack pixels, thereby strengthening the weight of the classification loss of crack pixels in the training label and weakening the weight of the classification loss of noncrack pixels. According to this strategy, the sample weight is reset for this extremely unbalanced sample. The proposed side network loss function is defined as follows:

$$L_{side}^{(m)}(X, Y, W, w^{(m)}) = -\beta \sum_{i \in Y_+} \log P_i(y_i = 1 | X; W, w^{(m)}) - (1 - \beta) \sum_{i \in Y_-} \log P_i(y_i = 0 | X; W, w^{(m)}), \quad (8)$$

where β is a balance factor. Let $\beta = |Y_-| / |Y|$ and $1 - \beta = |Y_+| / |Y|$, respectively, where $|Y_-|$ and $|Y_+|$ are the total numbers of pixels in the crack image Y^n and the input image X^n in the truth map, respectively. That is, β represents the ratio of crack pixels to the total number of pixels in the truth map Y^n , $(1 - \beta)$ represents the ratio of noncrack pixels to the total number of pixels in the truth map Y^n , and $P_i(y_i | X, W, w^{(m)})$ represents the probability value predicted by the m th side network using the sigmoid function at the i th pixel.

3.2 | Fusion layer loss

In the network fusion layer, the network output of each side obtains the response $\hat{Y}_{fuse}$ of the fusion layer network through logistic regression, and then these responses are connected for effective fusion. The training acquires the feature fusion weight h from the network output results of each stage. The loss function $L_{fuse}(Y, W, w, h)$ of the network fusion layer is defined as

$$L_{fuse}(Y, W, w, h) = DS(Y, \hat{Y}_{fuse}), \quad (9)$$

where $\hat{Y}_{fuse} \equiv \delta\left(\sum_{m=1}^M h_m \hat{A}_{side}^{(m)}\right)$ represents the predicted value output by the fusion layer network, and $h = (h_1, \dots, h_M)$ represents the feature fusion weights at different stages. $\hat{A}_{side}^{(m)} \equiv \{a_i^{(m)}, i = 1, \dots, |Y|\}$ represents the output response of the m th stage network, $|Y|$ indicates the total number of pixels in the input image, and $DS(\cdot)$ represents the cross-entropy loss between the predicted value $\hat{Y}_{fuse}$ of the fusion layer and the true value Y of the input crack image.

3.3 | Overall network loss

The overall network loss is defined as the sum of the side network loss $L_{side}(\cdot)$ and the fusion layer loss $L_{fuse}(\cdot)$. The training of the proposed network uses gradient descent to iteratively optimize the network parameter W , the weight parameter w of each connection layer and the weight parameter h of the different feature fusion stages, which effectively limits the predicted output of the side cracks and improves the ability of the network to extract linear structures. The optimization process is defined as follows:

$$L^*(X, Y, W, w, h) = \operatorname{argmin}(L_{side}(X, Y, W, w) + L_{fuse}(Y, W, w, h)), \quad (10)$$

where $L_{side}(\cdot)$ denotes the side network loss of the holistically nested network and $L_{fuse}(\cdot)$ represents the loss of the fusion layer of the side network.

4 | EXPERIMENTAL RESULTS AND ANALYSIS

4.1 | Experimental platform configuration and dataset construction

The experimental platform hardware configuration consists of an Intel Xeon E5 2650 processor with 32 GB of memory and 2 NVIDIA TitanV Volta 12 GB graphics cards (GPU processing units). The development environment is a Spyder integrated development environment and Python 3.6. The operating system is Windows 10. The deep learning framework uses PyTorch 1.1, Nvidia's CUDA8.0 GPU computing platform and the cuDNN6.0 deep learning GPU acceleration library. The experimental dataset includes the public Bridge_Crack_Image_Data³⁵ and CFD⁵² datasets. Bridge_Crack_Image_Data contains a total of 2000 bridge crack images with a size of 1024×1024 . This experiment adopts 1800 images as a training set and uses the remaining 200 images as a test set. Because the background region is too large, in order to improve the network's ability to learn crack features and reduce invalid feature learning, the training set data are enhanced, and the crack region in each image is selected, cropped, rotated, and flipped, generating 28,818 images with a size of 256×256 . The CFD dataset includes 118 pavement crack images sized 480×360 . For this study, we selected 100 as a training set and used the remaining 18 as a test set. These training set data are also enhanced to strengthen the feature learning of the crack regions through cropping, rotating and flipping. Ultimately, 2000 crack images with a size of 360×360 are generated.

To verify the effectiveness of the proposed CDFHNet network for crack segmentation, qualitative and quantitative comparison experiments were carried out with HED,⁵¹ FCN,⁵³ SegNet,⁵⁴ UNet,⁵⁵ and RCF.⁵⁶ All of these networks use a VGG-16 backbone network. UNet and SegNet use deconvolution to reconstruct the features of each stage, while HED and RCF use a holistic nested network. In the experiment, the batch size is 15, the epoch number is 30, the initial learning rate is 0.01, and the momentum is 0.85. The experiment uses an exponential decay, which decays every 5 epochs, and the decay coefficient is 0.7. The training loss differences among FCN, HED, CRF, SegNet, UNet and CDFHNet are shown in Figures 7 and 8. Figure 7 shows the training loss after the 25th iteration, and Figure 8 shows the training loss after the 40th iteration, at which point the loss curves of all networks have basically converged. The FCN, HED, and CRF networks have larger losses, and SegNet, UNet, and CDFHNet have smaller losses. The loss value of the proposed CDFHNet network is the smallest.

The comparison experiment is divided into four parts. The first two parts are feature activation map visualization experiments which are divided into two sets: the first set verifies the influence of the feature fusion module on the activation map, and the second set verifies the influence of the holistically nested network module on the activation map. The third and fourth parts compare the performance differences between the proposed network and the other networks based on the Bridge_Crack_Image_Data and CFD datasets, respectively.

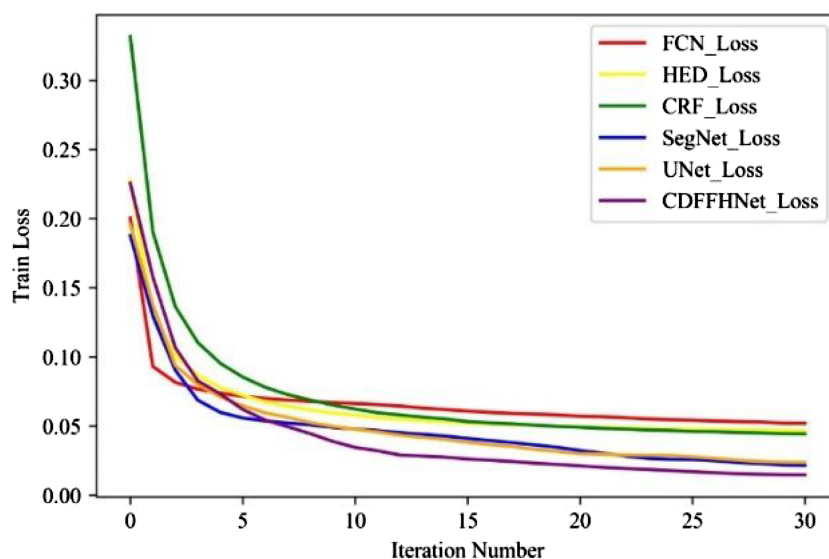


FIGURE 7 The training loss differences among the compared networks based on the Bridge_Crack_Image_Data dataset

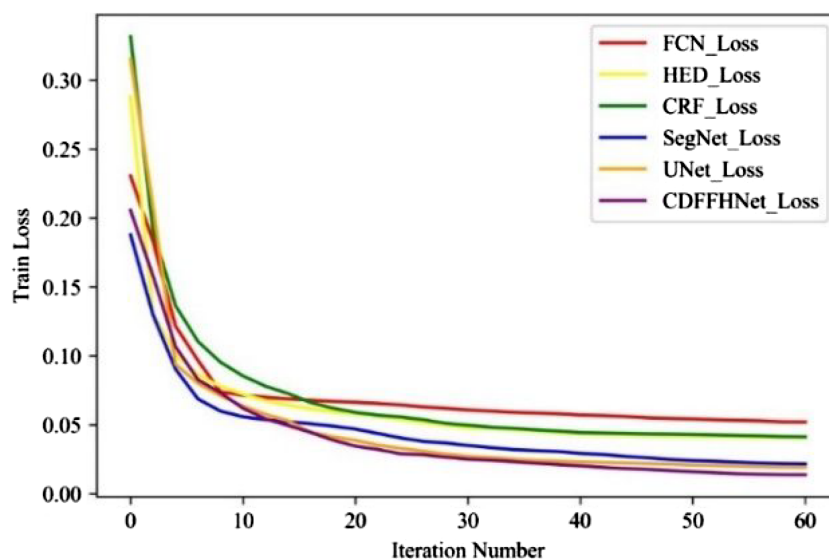


FIGURE 8 The training loss differences among the compared networks based on the CFD dataset

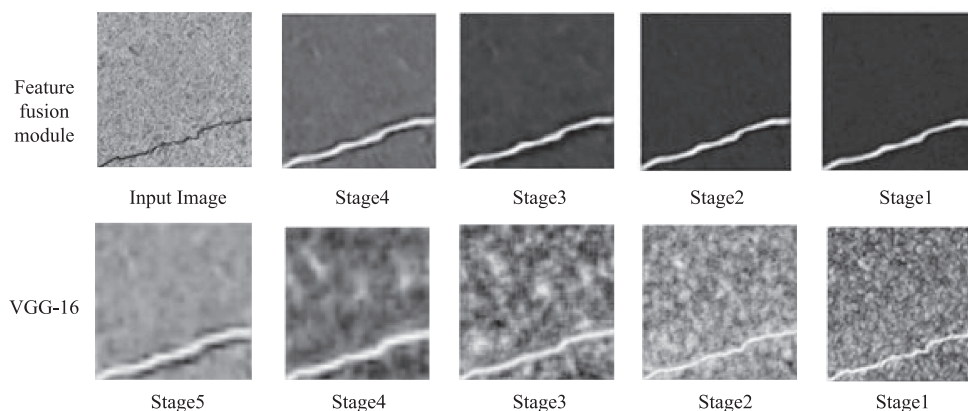


FIGURE 9 Visual comparison between VGG-16 and feature fusion module feature maps

The quantitative analysis adopts the accuracy rate, recall rate, and average F1-score as evaluation metrics. For crack pixel classification, there are only two classes: crack and noncrack. It is assumed that crack pixels are positive samples and that noncrack pixels are negative samples.

4.2 | Feature activation map visualization experiment

To verify the effectiveness of the feature fusion module and the holistically nested network module for the activation map, we conducted two sets of visualization experiments for verification.

4.2.1 | Visualization experiment on the effectiveness of the feature fusion module

To verify the performance of the feature fusion module in extracting the features of cracks at each stage, we visualize the activation map of each stage of the VGG-16 network and the activation map of each stage of the VGG-16 network based on the feature fusion module. In Figure 9, the first row presents the visualized activation map using the feature fusion module, and the second row includes the visualized activation map from each stage of the VGG-16 network. Figure 9 shows that crack features are captured at each stage of the VGG-16 network. The features in the lower stage

not only contain more detailed information, such as local and pixel-level features of cracks, but also strong interference from background texture and noise, especially in stage 1, stage 2, and stage 3. In contrast, the features in the higher stage contain more semantic features but less noise and background interference features, especially in stage 4 and stage 5. The visualized VGG-16 network activation map based on the feature fusion module is highly identifiable in both crack and noncrack regions. This result appears because the fifth-stage features containing more semantic crack information of the VGG-16 network are used for deconvolution learning. Therefore, the deconvolution at each stage of the reconstruction effectively eliminates interference from noise and background texture features, while the convolution and fusion operations at the same stage of VGG-16 largely eliminate interference from noise and background texture features. In addition, the feature fusion module fuses the higher stage image high-level semantic information with the low-level crack detail feature information, thereby significantly increasing low-level deep crack semantic information to further increase the representation ability of features.

4.2.2 | Visualization experiment of holistically nested network effectiveness

To verify the effectiveness of the holistically nested network, we visualized the activation maps of the prediction results of the proposed CDFHNet, SegNet, and UNet, as shown in Figure 10. The proposed CDFHNet, UNet, and SegNet all use a VGG-16 as the backbone network and perform deconvolution to reconstruct the features at each stage. However, UNet and SegNet do not use a holistically nested network; they output the prediction results at the top level and do not merge the outputs of the other stages. As the visualization results show, the UNet and SegNet networks still contain noise in the noncrack regions: this is especially true for SegNet, which is greatly affected by noise. In contrast, the proposed CDFHNet effectively enhances the feature response of the crack region and weakens the feature response of the noncrack region through the feature fusion operation at each stage, thereby effectively improving the network's response to linear crack structural features. The predicted result has a clear background, and the crack region segmentation effect is good.

4.3 | The contribution of each stage of the holistically nested network module

To verify the contribution of each stage of the holistically nested network to the experimental results, we separately removed stages 1–4 while retaining the remaining stages and performed comparative tests. These experimental results are shown in Table 3. The removal of stage 1 shows that it contributes the most to the results. On the

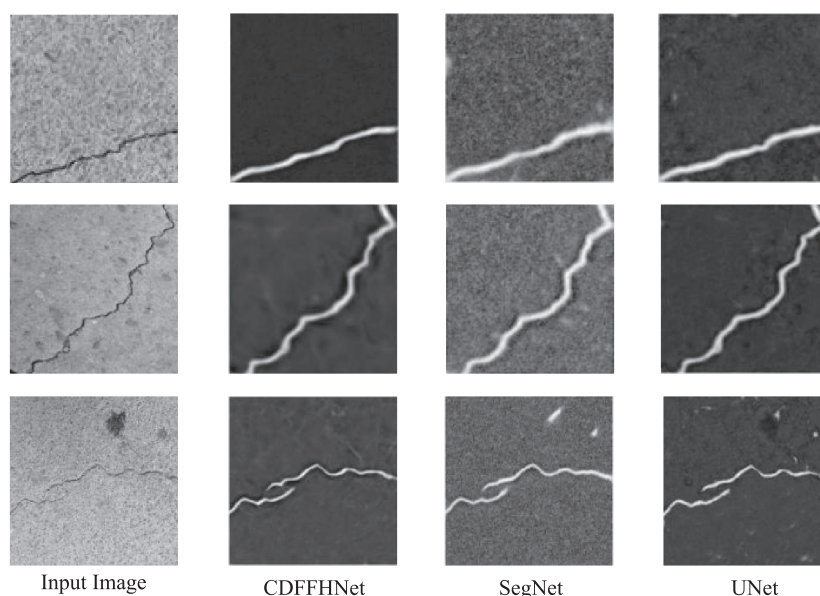


FIGURE 10 Visualization comparison of holistically nested network effectiveness

TABLE 3 The contribution of each stage of the holistically nested network module

	Bridge_Crack_Image_Data			CFD		
	P	R	F1-score	P	R	F1-score
CDFHNet w/o stage1	0.8534	0.6907	0.7634	0.8765	0.6842	0.7685
CDFHNet w/o stage2	0.8756	0.6913	0.7726	0.8805	0.6869	0.7717
CDFHNet w/o stage3	0.8942	0.6945	0.7817	0.9058	0.6939	0.7858
CDFHNet w/o stage4	0.9134	0.7011	0.7932	0.9211	0.7033	0.7975
CDFHNet	0.9156	0.7063	0.7974	0.9265	0.7098	0.8080

Note: The best result is highlighted in bold font.

Bridge_Crack_Image_Data dataset,³⁵ the accuracy rate fell by 6.22%, the recall rate fell by 1.56%, and the F1-score fell by 3.4%. Similarly, on the CFD dataset, the accuracy rate fell by 5%, the recall rate fell by 3.23%, and the F-score fell by 3.95%. The removal of stage 4 shows that it makes the smallest contribution to the result. On the Bridge_Crack_Image_Data dataset, the accuracy rate dropped by 0.22%, the recall rate dropped by 0.52%, and the F1-score dropped by 0.42%. On the CFD dataset, the accuracy rate dropped by 0.54%, the recall rate dropped by 0.65%, and the F1-score dropped by 1.05%. As Table 3 shows, all features from every stage contribute to the result, indicating that fusing the convolutional features at different stages effectively improves the network crack detection performance.

4.4 | Analysis of network comparison experiment results

4.4.1 | Bridge_Crack_Image_Data dataset comparison experiment

To verify the effectiveness of the proposed CDFHNet for crack segmentation, we use the proposed CDFHNet and benchmark HED,⁵¹ FCN,⁵³ SegNet,⁵⁴ UNet,⁵⁵ and RCF⁵⁶ networks to perform crack segmentation comparison experiments on the Bridge_Crack_Image_Data dataset. Partial test results of the Bridge_Crack_Image_Data dataset are shown in Figure 11. The first column in Figure 11 shows the original image, the second column shows the ground truth, and the third to seventh columns show the detection results of the FCN, UNet, SegNet, HED, and RCF networks, respectively. The last column shows the detection results of the proposed CDFHNet network.

Qualitative analysis

As the comparison result graph shows, for the test images with less background interference and clear cracks (e.g., test images 1 and 2 in Figure 11a), all the tested networks clearly detect the cracks. However, there are concrete stains on test images with more complex backgrounds and more interference (e.g., the test images of labels 3 and 4 in Figure 11a, where the background is a gravel road), but the cracks are generally clear. The comparison results show that FCN misjudges stained regions as crack regions. A fracture occurred in the unclear region of the crack, and crack information could not be fully detected. In comparison with FCN, SegNet and UNet use deconvolution to reconstruct the feature maps of each stage and use a combination of deep semantic information and shallow detail information to better detect crack regions. However, these networks misjudge some large stains. HED and RCF networks distinguish crack and noncrack regions separately from multiple-stage features, thereby enhancing the feature response of the crack region and weakening the feature response of the noncrack region and obtaining clearer crack segmentation results. However, these networks still misjudge some stains, and the detected crack region is discontinuous. This result occurs because the features of each stage of HED and RCF are derived directly from VGG-16, and these individual stage characteristics are not reconstructed by deconvolution: they therefore lack semantic information, causing fractures to occur in low-contrast crack regions. The proposed CDFHNet network uses a feature fusion module to enhance the crack features of each stage. Holistically nested network suppresses the feature responses of noncrack regions and optimizes the responses of the crack feature activation map. Therefore, not only are cracks completely extracted, but the stain interference region is also correctly segmented. For scenes with less background interference, where the crack region and background are small and cracks are relatively minor, as shown in test images 5, 6, 7, 8, and 10 in Figure 11a, FCN lacks shallow details and therefore misses cracks in many regions. UNet and SegNet integrate shallow and deep

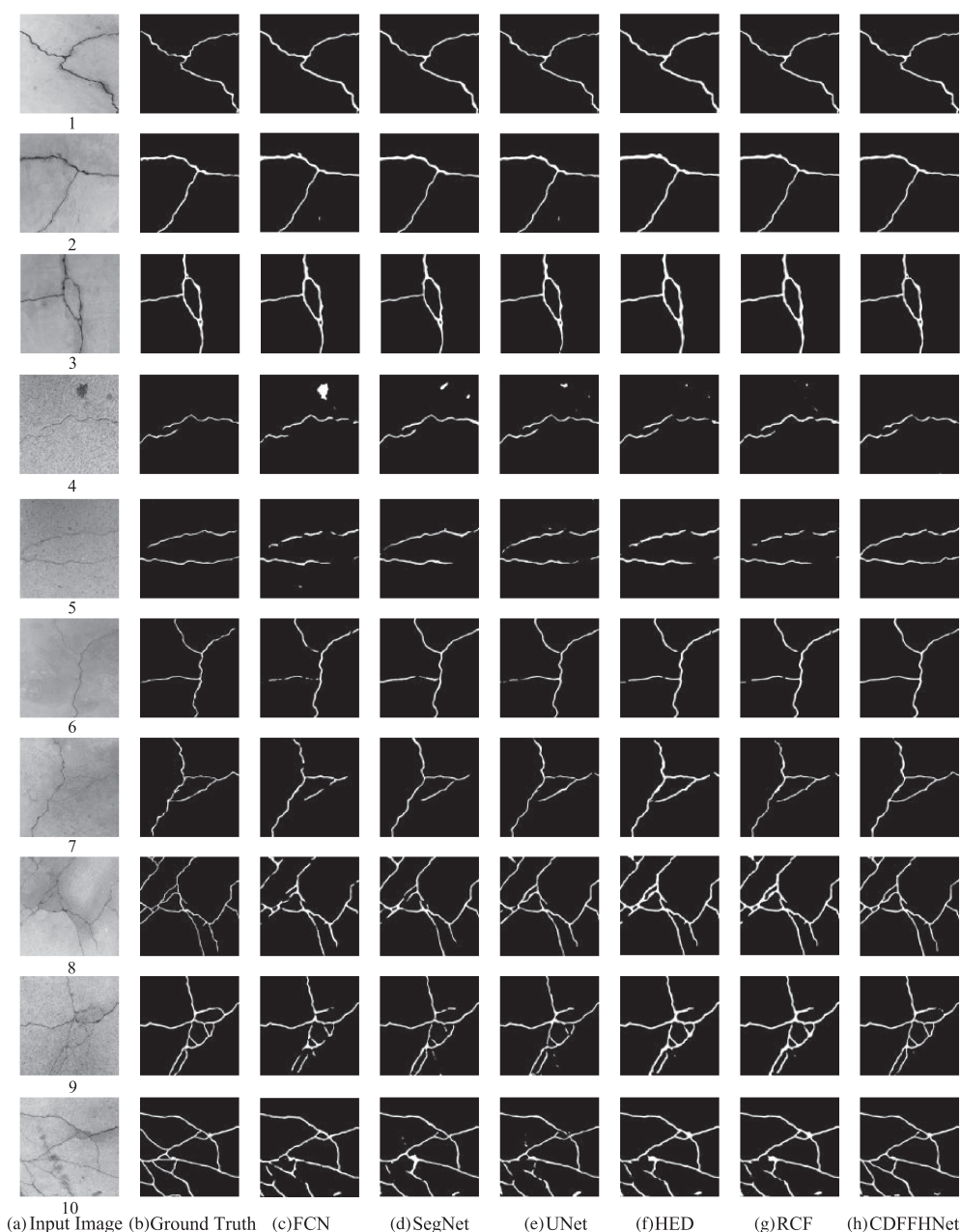


FIGURE 11 Comparison of partial test results on the Bridge_Crack_Image_Data dataset

information to improve the detection of such cracks. RCF and HED directly use the features extracted by VGG-16 without feature enhancement and directly use the multiscale integral nested network; as a result, cracks cannot be detected well either. The proposed CDFFHNet network integrates contextual information and can not only accurately extract small cracks but also obtain a continuous crack structure, showing that the proposed network has a strong ability to capture the features of the linear topological structure of cracks.

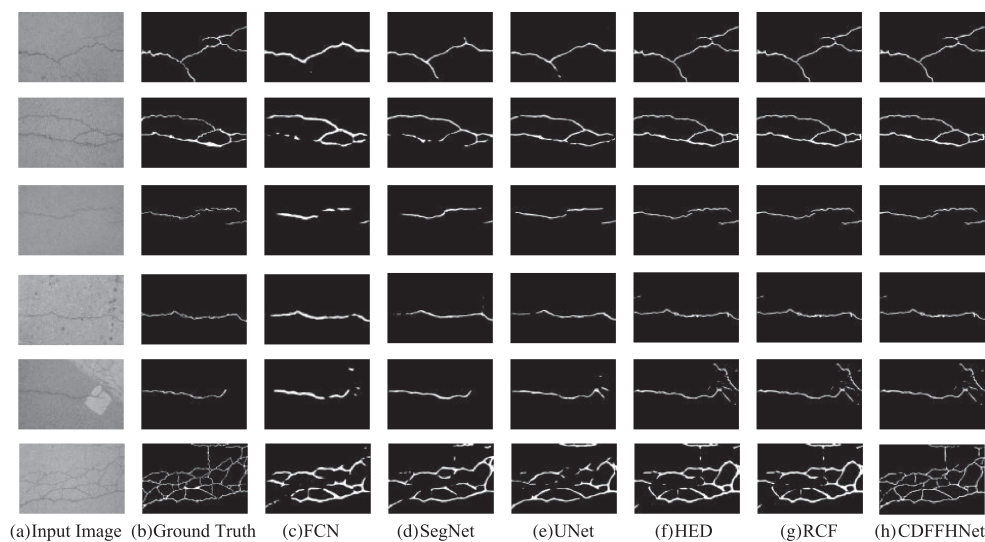
Quantitative analysis

The quantitative analysis of the detection results of different networks on the Bridge_Crack_Image_Data dataset is shown in Table 4 as indicated by accuracy, recall, and F1-scores. As shown in Table 4, the F1-scores of UNet and SegNet are better than those of HED, RCF and FCN, indicating that using the features of deconvolution reconstruction at each stage plays an important role in crack extraction. However, the accuracy scores of HED and RCF are higher than those of UNet, SegNet, and FCN. This result occurs because HED and RCF use a holistically nested network to

TABLE 4 Quantitative analysis and comparison of the Bridge_Crack_Image_Data dataset

Network	Precision	Recall	F1-score	Standard deviation		FPS	Parameters size
				Precision	F1-score		
HED	0.9167	0.6503	0.7608	0.0876	0.0797	6.57	63.9M
FCN	0.8784	0.6816	0.7679	0.0983	0.0831	6.75	106M
SegNet	0.8745	0.6887	0.7705	0.0557	0.0670	6.28	112M
UNet	0.8945	0.6822	0.7740	0.0561	0.0660	6.24	51.1M
RCF	0.9086	0.6607	0.7650	0.0865	0.0764	6.48	56.4M
CDFHNet	0.9156	0.7063	0.7974	0.0235	0.0427	6.20	69.1M
CDFHNet + CAM	0.9224	0.7101	0.8024	0.0233	0.0414	6.18	79.1M

Note: The best result is highlighted in bold font.

**FIGURE 12** Comparison of partial test results on CFD dataset

improve the extraction of low-contrast crack regions. The proposed CDFHNet network achieves better results than all comparison networks in all three metrics: accuracy, recall and F1-score. Compared with the proposed CDFHNet network, the CDFHNet + CAM network contains a CAM to simulate the features of the original VGG-16 crack features at each stage; therefore, this network improves the accuracy, recall, and F1-scores by 0.68%, 0.38% and 0.50%, respectively. Compared with the FCN-8s, HED, RCF, UNet, and SegNet networks, CDFHNet + CAM improves the F1-scores by 3.45%, 4.16%, 3.74%, 2.84%, and 3.19%, respectively.

To better prove the differences between the proposed CDFHNet network and other benchmark networks, we show the standard deviation of the accuracy and F1-score in Table 4. As shown in Table 4, compared with other networks, the standard deviations of precision and F1-score of the proposed CDFHNet and CDFHNet + CAM are the smallest, indicating that the proposed network is relatively robust. We also compare frames per second (FPS) and network parameter size with other networks. Since the proposed CDFHNet network is based on an encoding–decoding architecture, the decoding process results in more computing time for image crack segmentation; thus, its speed is not the fastest, but it is closer to the other networks. The FCN, HED, and RCF networks speeds are faster since they have no feature decoding structure, but their accuracy is lower. The parameter sizes are listed in Table 4: the parameter size of the proposed CDFHNet network is moderate. Therefore, based on the comparison of the qualitative indicators, the proposed CDFHNet network has better recognition ability and robustness.

4.4.2 | CFD dataset comparison experiment

To further verify the effectiveness and robustness of the proposed CDFHNet network, we conducted a comparative crack detection experiment on the CFD dataset, and the results are shown in Figure 12. The first column in Figure 12 is the original image, the second column is the ground truth, the third to seventh columns show the detection results of the FCN, UNet, SegNet, HED, and RCF networks, respectively, and the last column shows the detection results of the proposed CDFHNet network.

Qualitative analysis

CFD is a road crack dataset in which crack regions are inconspicuous and small. As shown in Figure 12, the proposed CDFHNet network detection results are best compared with the detection results of FCN, UNet, SegNet, HED, and RCF for crack areas with different structures. In labels 1–3 of Figure 12a and some regions with inconspicuous crack contrast and small cracks, the results of other networks include missed detections and false detections. However, the proposed network still detects cracks in these difficult images and retains crack details. In labels 4–6 of Figure 12a, oil droplet interference occurs in the images, but the proposed network can filter out the oil droplet interference and clearly extract the crack region. UNet and SegNet mistake some oil droplets for cracks. Traffic sign lines occur in label 5, and there are thin linear-shaped cracks on the traffic signs. The proposed network detects these linear cracks very well, indicating that the proposed network has a strong ability to capture linear structural features. Label 6 is a large-scale crack image, and some crack areas have low contrast. The detection ability of other networks is weaker, and there are more missed detections. However, the proposed network correctly detects more than 90% of the region and retains crack details. There are no missed cracks, and even in some difficult-to-distinguish crack structures, the proposed CDFHNet network detects cracks with linear structures, again indicating that the proposed network captures linear topology structural features well.

Quantitative analysis

The quantitative analysis of the detection results of different networks in the CFD dataset is shown in Table 5, using accuracy, recall, and F1-scores for evaluation. As shown in Table 5, the accuracy, recall, and F1-score values of the proposed CDFHNet network reach 0.9196, 0.9098 and 0.9147, respectively, and are consistently better than those of the comparison networks. Compared with FCN-8S, HED, RCF, UNet, and SegNet, the proposed CDFHNet + CAM network increases the F1-score by 6.32%, 4.7%, 4.47%, 2.93%, and 3.45%, respectively, indicating the superior recognition capabilities of the proposed network. As shown in Table 5, compared with other networks, the standard deviations of precision and F1-score of the proposed CDFHNet are the smallest, which further shows that the proposed network is relatively robust.

The comparative experiments and quantitative analysis on Bridge_Crack_Image_Data and CFD datasets show that the proposed CDFHNet network effectively ameliorates the interference with crack detection results from factors such as complex crack backgrounds, small differences in background and foreground, and linear and slender cracks that readily break during segmentation, and can accurately segment concrete crack regions. Therefore, the proposed network can not only accurately detect concrete cracks and better identify crack linear topology structural features but is also robust to various strong interferences under complex backgrounds.

TABLE 5 Quantitative analysis and comparison based on CFD dataset

Network	Precision	Recall	F1-score	Standard deviation		FPS	Parameters size
				Precision	F1-score		
HED	0.8656	0.8699	0.8677	0.0843	0.0786	40.57	63.9M
FCN	0.8564	0.8467	0.8515	0.1065	0.0897	40.82	106M
SegNet	0.8816	0.8789	0.8802	0.0698	0.0645	39.22	112M
UNet	0.8811	0.8897	0.8854	0.0697	0.0724	39.17	51.1M
RCF	0.8698	0.8703	0.8700	0.0807	0.0788	40.45	56.4M
CDFHNet	0.9107	0.9022	0.9064	0.0313	0.0387	38.85	69.1M
CDFHNet + CAM	0.9196	0.9098	0.9147	0.0342	0.0405	38.71	79.1M

Note: The best result is highlighted in bold font.

5 | CONCLUSION

This research proposed a concrete crack segmentation method based on convolution–deconvolution feature fusion with holistically nested networks. The proposed network uses channel attention to screen the original features extracted by the VGG-16 network to simulate an effective crack response and avoid interference from invalid and incorrect feature responses. Then, through convolution–deconvolution feature fusion, deep semantic information of deconvolution and shallow features of convolution are effectively fused, which improves the quality of crack features. Finally, a holistically nested network is used to achieve multiscale convolution fusion, which enhances the responses to crack region features, weakens the responses to noncrack region features, and improves the network's ability to capture linear topological structural features. Feature activation visualization proves that the proposed CDFHNet network has strong linear topology structure capture ability. Many experimental results show that the proposed network offers better recognition ability and stronger robustness.

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AUTHOR CONTRIBUTIONS

Shengjun Xu and Ming Hao: Study concept and design. **Shengjun Xu, Ming Hao and Guanghui Liu:** Drafting of the manuscript. **Yuebo Meng:** Statistical analysis. **Jiuqiang Han and Ya Shi:** Study supervision.

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